

DATA FLOW DIAGRAM & USER STORIES

Date	30 September 2026
Team ID	SWTID-2026-3024
Project Name	AI-Augmented Backend Development AI BlogNest API
Maximum Marks	4 marks

1. DATA FLOW DIAGRAM

Data Flow Diagram – AI BlogNest

A Data Flow Diagram represents how information moves through the AI BlogNest system. It shows how data is received from users, processed by the backend application, stored in the database, sent to the AI service when required, and returned to the user as a response.

The main external entities and system interactions are:

1. User / Content Creator

- Sends registration and login information.
- Sends blog creation and management requests.
- Sends comments and content-related requests.
- Provides prompts for AI content generation.
- Requests content summarization.
- Receives API responses and generated content.

2. Administrator

- Manages users and roles.
- Controls authorized system operations.
- Monitors and manages content-related activities.

3. AI Service

- Receives content-generation prompts from the backend.
- Processes AI requests.
- Returns generated blog content or summaries to the backend.

4. MongoDB Database

- Stores user information.
- Stores blog posts.
- Stores comments.

- Stores categories, tags, and related application data.

Main Data Flow



Process 1: User Authentication

- Receives registration and login details.
- Validates user information.
- Hashes passwords using bcrypt.js.
- Generates JWT authentication tokens.
- Stores and retrieves user information from MongoDB.

Process 2: User Authorization

- Verifies JWT tokens.
- Identifies the user's role.
- Allows or denies access according to permissions.

Process 3: Blog Management

- Creates blog posts.
- Retrieves blog posts.
- Updates blog posts.
- Deletes blog posts.
- Manages categories and tags.

Process 4: Comment Management

- Receives comments.
- Associates comments with blog posts.
- Retrieves and manages comments.

Process 5: AI Content Generation

- Receives a topic or prompt.
- Sends the request to the AI service.

- Receives generated content.
- Returns the generated content to the user.

Process 6: AI Content Summarization

- Receives existing blog content.
- Sends the content to the AI service.
- Receives a concise summary.
- Returns the summary through the API.

Process 7: Database Management

- Stores and retrieves users, blogs, comments, categories, and tags using MongoDB and Mongoose.

2. USER STORIES

The following user stories define the major functional requirements of the AI BlogNest system. Each story describes the expected functionality from the perspective of the user or system administrator.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Content Creator	Registration & Authentication	USN-1	As a user, I can register for AI BlogNest by providing the required details.	User can submit valid details and a new account is created successfully.	High	Sprint-1
Content Creator	Registration & Authentication	USN-2	As a registered user, I can log into the application using my email and password.	Valid credentials allow successful login and authentication information is returned.	High	Sprint-1
Content Creator	Registration & Authentication	USN-3	As a user, my password should be securely stored using password hashing.	Password is not stored as plain text and is securely hashed.	High	Sprint-1
Content Creator	Authentication	USN-4	As an authenticated	A valid JWT token is	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
			user, I can receive a JWT token for accessing protected APIs.	generated after successful authentication.		
Administrator	Role Management	USN-5	As an administrator, I can manage user roles and permissions.	Users are assigned appropriate roles and restricted from unauthorized operations.	High	Sprint-1
User	Security & Validation	USN-6	As a user, I should receive an appropriate response when invalid registration or login data is submitted.	Invalid or incomplete data is rejected with a meaningful validation message.	High	Sprint-1
User	Security & Validation	USN-7	As a user, I should not be able to access protected APIs without valid authentication.	Unauthorized requests are rejected		